NETWORK CABLES – STUDY AND FABRICATION OF ETHERNET CABLE

AIM

To examine the major types of network cables, their features, uses, benefits, and limitations, and to construct an Ethernet straight-through or cross-over cable using RJ45 connectors.

a) STUDY OF DIFFERENT TYPES OF NETWORK CABLES

The commonly used networking cable types considered in this experiment are:

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Shielded and Screened Twisted Pair (SSTP) cable
4. Coaxial cable
5. Fibre optic cable

1. Unshielded Twisted Pair (UTP) Cable

Category	Maximum Data Transmission	Advantages	Disadvantages	Application / Use
Category 3	10 Mbps	Low cost, simple installation, and relatively small cable diameter.	More vulnerable to electromagnetic interference (EMI) and noise.	10Base-T Ethernet
Category 5	Up to 100 Mbps	Economical and easy to install.	More prone to interference than shielded cable types.	Fast Ethernet, Gigabit Ethernet
Category 5e	Up to 1 Gbps	Affordable, easy to install, and suitable for high-speed networking.	Provides less interference protection.	Fast Ethernet, Gigabit Ethernet

2. Shielded Twisted Pair (STP) Cable

Category	Maximum Data Transmission	Advantages	Disadvantages	Application / Use
Category 6 / 6a	Up to 10 Gbps	Shielding reduces noise and interference and provides higher performance than UTP.	More expensive and requires additional installation effort.	Gigabit Ethernet, 10G Ethernet over suitable distances, data centres

3. Shielded and Screened Twisted Pair (SSTP) Cable

Category	Maximum Data Transmission	Advantages	Disadvantages	Application / Use
Category 7	Up to 10 Gbps	Provides excellent shielding and minimizes	Costlier and comparatively harder to install.	Gigabit Ethernet and 10G Ethernet up to 100 m

		interference.		

4. Coaxial Cable

Category	Maximum Data Transmission	Advantages	Disadvantages	Application / Use
RG-6, RG-59, RG-11	Approximately 10–100 Mbps	Good bandwidth, strong resistance to interference, low signal loss, and versatile use.	Limited transmission distance, relatively expensive, and physically bulky.	Television networks and high-speed Internet connections

5. Fibre Optic Cable

Category	Maximum Data Transmission	Advantages	Disadvantages	Application / Use
Single Mode / Multi Mode	Up to 100 Gbps and beyond, depending on equipment	Very high speed and bandwidth, strong security, and long-distance transmission capability.	Expensive and needs skilled installation.	Long-distance communication, data centres, and high-speed networks

b) CONSTRUCTION OF AN ETHERNET CROSS-OVER / STRAIGHT-THROUGH CABLE

Tools and Parts Required

- Ethernet cable (CAT5/CAT5e)
- Crimping tool
- Two RJ45 plugs
- Cable stripper/cutter
- RJ45 plug shields (optional)
- T568A / T568B wiring reference

Difference Between Straight-Through and Cross-Over Cable

Straight-Through Cable

A straight-through cable follows the same wiring standard on both ends. The possible arrangements are T568A–T568A or T568B–T568B.

Typical connections include:

- PC to Switch
- PC to Hub
- Router to Switch

Cross-Over Cable

A cross-over cable uses different wiring standards at its two ends, such as T568A on one side and T568B on the other. It has traditionally been used for connecting similar devices.

- PC to PC
- Switch to Switch
- Router to Router

Many modern networking devices include Auto-MDI/MDIX, allowing them to automatically recognize the required cable arrangement.

Procedure

Step 1: If cable shields are being used, slide them onto the Ethernet cable before fitting the RJ45 connectors.

Step 2: Remove about 1.5 cm of the outer jacket from each cable end using a cable stripper or suitable crimping tool.

Step 3: Separate and untwist the four wire pairs. Arrange the conductors according to the selected T568A or T568B standard. Use the same standard at both ends for a straight-through cable, and use T568A at one end and T568B at the other for a cross-over cable.

Step 4: Place the wires in the required sequence, straighten them, and trim them evenly so that their ends are aligned.

Step 5: With the RJ45 plug held so that the gold contacts face you, insert the conductors carefully while maintaining the chosen order.

Step 6: Push the conductors fully into the RJ45 plug. Make sure a small portion of the outer jacket enters the connector so the cable is properly secured during crimping.

Step 7: Place the RJ45 plug into the crimping tool and press firmly to lock the conductors and contacts in position.

Step 8: Repeat the termination process at the opposite end, following the correct wiring standard for the cable type.

Step 9: Check the finished cable with a LAN cable tester, or connect it to suitable network devices to verify that it works correctly.